Spin part 2

Let $|s\rangle$ be a spin with polar angle θ and azimuth angle ϕ.

$$|s\rangle = \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2)\, e^{i\phi} \end{pmatrix}$$

Spin measurement probabilities are the transition probabilities from $|s\rangle$ to an eigenstate.

The eigenstates are given by the eigenvectors

$$|x_+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} \qquad |y_+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ i \end{pmatrix} \qquad |z_+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$|x_-\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix} \qquad |y_-\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -i \end{pmatrix} \qquad |z_-\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Hence in the x direction

$$\Pr(+) = |\langle x_+|s\rangle|^2 = \tfrac{1}{2} + \tfrac{1}{2}\sin\theta\cos\phi$$
$$\Pr(-) = |\langle x_-|s\rangle|^2 = \tfrac{1}{2} - \tfrac{1}{2}\sin\theta\cos\phi$$

In the y direction

$$\Pr(+) = |\langle y_+|s\rangle|^2 = \tfrac{1}{2} + \tfrac{1}{2}\sin\theta\sin\phi$$
$$\Pr(-) = |\langle y_-|s\rangle|^2 = \tfrac{1}{2} - \tfrac{1}{2}\sin\theta\sin\phi$$

In the z direction

$$\Pr(+) = |\langle z_+|s\rangle|^2 = \tfrac{1}{2} + \tfrac{1}{2}\cos\theta$$
$$\Pr(-) = |\langle z_-|s\rangle|^2 = \tfrac{1}{2} - \tfrac{1}{2}\cos\theta$$

For each direction we have $\Pr(+) + \Pr(-) = 1$ as required by total probability.

Eigenmath script